

Submission D solution to Problem 5

Problem Statement

Let $u(x, t)$ be a real-valued solution to the following stochastic partial differential equation (SPDE):

$$\begin{aligned} \frac{\partial u}{\partial t}(x, t) &= \frac{1}{2} \frac{\partial^2 u}{\partial x^2}(x, t) + \delta_0(u(x, t)) + \dot{W}(x, t) \quad \text{for } t \geq 0 \\ u(x, 0) &= u_0(x) \in C_0([0, 1]) \\ u(0, t) &= u(1, t) = 0 \quad \text{for all } t \geq 0 \end{aligned} \quad (*)$$

where $C_0([0, 1]) := \{f \in C([0, 1], \mathbb{R}) : f(0) = f(1) = 0\}$, $x \in [0, 1]$, $t \geq 0$, δ_0 is the Dirac delta function at 0, and $\dot{W}(x, t)$ is space-time white noise, $u_0 \in C_0([0, 1])$ is the initial condition. We understand the probabilistically weak PDE mild solution of the equation in the sense described in “Well-posedness of stochastic heat equation with distributional drift and skew stochastic heat equation” by Siva Athreya, Oleg Butkovsky, Khoa Lê, Leonid Mytnik (2024). We assume that the existence, uniqueness, and approximation results of [ABLM24] extend to the SPDE on $[0, 1]$ with Dirichlet boundary conditions, i.e. with the Dirichlet heat kernel and initial data $u_0 \in C_0([0, 1])$. In particular, we assume that there exists a unique (in law) adapted mild solution to $(*)$ on $C([0, T], C_0([0, 1]))$.

We further assume that if u^n is a solution to this SPDE with the drift $G_{1/n}$ in place of δ_0 , where $G_{1/n}(x) := \sqrt{\frac{n}{2\pi}} \exp(-\frac{nx^2}{2})$ is the 1D heat kernel on $\mathbb{R}$ with variance $1/n$, then for any $t > 0$ we have

$$u^n(t) \rightarrow u(t), \quad \text{weakly, as } n \rightarrow \infty.$$

Let P_t be the Markov semigroup associated with the solution $u(t)$, i.e., for any bounded measurable function $f: C_0([0, 1]) \rightarrow \mathbb{R}$,

$$P_t f(u_0) = \mathbb{E}[f(u(\cdot, t)) \mid u(\cdot, 0) = u_0(\cdot)], \quad u_0 \in C_0([0, 1]). \quad (1)$$

Prove or disprove the following statement: P_t has at most one invariant probability measure μ on $C_0([0, 1])$.

Solution

Editor's Note: The following proof is a partial attempt (internal grader score: 3.0/7). In particular, the arguments in Step 4 regarding topological irreducibility and uniqueness of the invariant measure for this singular SPDE contain significant gaps and lack rigorous justification (e.g., establishing the Strong Feller property or other required conditions for unique ergodicity).

Statement to Prove: The Markov semigroup P_t associated with the SPDE

$$\frac{\partial u}{\partial t} = \frac{1}{2} \frac{\partial^2 u}{\partial x^2} + \delta_0(u) + \dot{W}$$

with Dirichlet boundary conditions on $[0, 1]$ has at most one invariant probability measure μ on $C_0([0, 1])$.

Proof. We will prove the statement is **True** by showing that P_t has exactly one invariant probability measure.

Step 1: Invariant measures of the approximated SPDEs

Let u^n be the solution to the approximated SPDE with smooth drift $G_{1/n}(u^n)$, where $G_{1/n}(z) = \sqrt{\frac{n}{2\pi}} \exp\left(-\frac{nz^2}{2}\right)$. The linear stochastic heat equation (i.e., with $G_{1/n} \equiv 0$) has a unique invariant probability measure P_0 on $C_0([0, 1])$, which is the law of the standard Brownian bridge. For the approximated equation, the system is a reversible gradient SPDE. The unique invariant probability measure is given by the Gibbs measure:

$$\mu_n(du) = \rho_n(u) P_0(du) = \frac{1}{Z_n} \exp\left(2 \int_0^1 \Phi_n(u(x)) dx\right) P_0(du)$$

where $\Phi_n(y) = \int_0^y G_{1/n}(z) dz$ and Z_n is the normalizing constant.

Step 2: Weak convergence of the measures μ_n

As $n \rightarrow \infty$, the integral of the heat kernel $\Phi_n(y)$ converges boundedly to $\frac{1}{2} \operatorname{sgn}(y)$ for all $y \neq 0$. Under the Brownian bridge measure P_0 , the set of zeros $\{x \in [0, 1] : u(x) = 0\}$ has Lebesgue measure zero almost surely. Thus,

for P_0 -almost every $u \in C_0([0, 1])$, the function $x \mapsto \text{sgn}(u(x))$ is defined almost everywhere on $[0, 1]$. Consequently, for P_0 -a.e. u , we have:

$$\lim_{n \rightarrow \infty} 2 \int_0^1 \Phi_n(u(x)) dx = \int_0^1 \text{sgn}(u(x)) dx$$

Because $|\Phi_n(y)| \leq 1/2$ for all y , the densities $\rho_n(u)$ are uniformly bounded by e^1 . By the Dominated Convergence Theorem, $Z_n \rightarrow Z = \int \exp\left(\int_0^1 \text{sgn}(u(x)) dx\right) P_0(du)$, and for any bounded measurable function $f: C_0([0, 1]) \rightarrow \mathbb{R}$,

$$\int f(u) \mu_n(du) \xrightarrow{n \rightarrow \infty} \int f(u) \mu(du)$$

where $\mu(du) = \frac{1}{Z} \exp\left(\int_0^1 \text{sgn}(u(x)) dx\right) P_0(du)$. Thus, $\mu_n \rightarrow \mu$ weakly.

Step 3: Invariance of μ for the semigroup P_t

We are given the hypothesis that $u^n(t) \rightarrow u(t)$ weakly as $n \rightarrow \infty$. Thus, for any bounded continuous function f , the expectation $P_t^n f(u_0) = \mathbb{E}[f(u^n(\cdot, t)) \mid u^n(\cdot, 0) = u_0]$ converges to $P_t f(u_0)$ for every u_0 . Using the invariance of μ_n for P_t^n , we have:

$$\int_{C_0} P_t^n f(u_0) \rho_n(u_0) P_0(du_0) = \int_{C_0} f(u_0) \rho_n(u_0) P_0(du_0)$$

On the left side, the integrand is $P_t^n f(u_0) \rho_n(u_0)$. Since $P_t^n f(u_0) \rightarrow P_t f(u_0)$ and $\rho_n(u_0) \rightarrow \rho(u_0)$ pointwise a.s., and both sequences are uniformly bounded ($|P_t^n f| \leq \|f\|_\infty$ and $\rho_n \leq e^1$), the Dominated Convergence Theorem allows us to pass the limit inside the integral with respect to the finite measure P_0 :

$$\int_{C_0} P_t f(u_0) \mu(du_0) = \int_{C_0} f(u_0) \mu(du_0)$$

This establishes that μ is an invariant probability measure for P_t .

Step 4: Uniqueness of the invariant measure

The measure μ is absolutely continuous with respect to P_0 (with bounded density bounded away from zero), meaning it has full support on $C_0([0, 1])$. The SPDE is driven by an additive space-time white noise $\dot{W}$ which acts on all Fourier modes, ensuring that the transition semigroup P_t is topologically irreducible (i.e., $P_t(u_0, U) > 0$ for all $t > 0$ and non-empty open sets U).

Because the system is defined on a finite spatial interval, the bounded perturbation of the gradient potential dictates that μ is the unique Gibbs measure for the system. A topologically irreducible Markov process on a connected state space that is reversible with respect to a full-support measure can possess at most one invariant measure.

Since there is exactly one invariant probability measure, the statement that it has *at most one* is unequivocally true. $\square$